



Color bleaching and oxygen diffusion in a niobium phosphate glass



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ABSTRACT

The color of niobium phosphate glasses ranges from clear to deep blue depending on their thermal history. In this study, a glass with the nominal composition (mole %) $23\text{K}_2\text{O} \cdot 40\text{Nb}_2\text{O}_5 \cdot 37\text{P}_2\text{O}_5$ was treated in oxidizing and reducing atmospheres at temperatures around T_g (730 °C) and the color at room temperature was analyzed using optical spectroscopy. Electron spin resonance (ESR) spectroscopy shows that the blue color is associated with the presence of Nb^{4+} ions, which oxidize to Nb^{5+} in the clear glasses. Bleaching kinetics were determined from a series of heat treatments below T_g in flowing oxygen, and analyzed with an diffusion model that indicated an activation energy of 150 ± 31 kJ/mol.

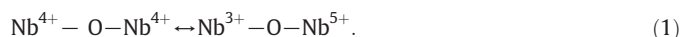
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1. Introduction

Niobium phosphate glasses are of interest for optical applications [1,2] and have been developed as hosts for radioactive wastes [3]. The color of niobium phosphate glasses varies from colorless [4,5] and yellowish [4] to dark blue [5], depending on the glass composition and preparation conditions. Chu et al. [5] observed that glasses from the $\text{SrO}-\text{Nb}_2\text{O}_5-\text{P}_2\text{O}_5$ system were blue when the $\text{P}_2\text{O}_5/\text{SrO}$ ratio was greater than one, and colorless when the ratio was less than one. Tawarayama et al. [6] found that initially clear alkali tungsten-niobium phosphate glasses turned dark blue when heated at temperatures close to T_g in reducing atmospheres, then were bleached to a colorless on subsequent heat treatments in air. This bleaching effect was attributed to W oxidation, from W^{5+} to W^{6+} , occasioned by the diffusion of hydrogen (dissolved into the glass during the reducing heat treatment) out of the glass.

A glass with the nominal molar composition $23\text{K}_2\text{O} \cdot 40\text{Nb}_2\text{O}_5 \cdot 37\text{P}_2\text{O}_5$ was developed to immobilize radioactive waste [3]. This glass possesses high chemical durability and stability against crystallization [7]. When melted in a microwave furnace at temperatures around 1200 °C, the resulting glass is yellowish, but when the same glass composition is melted at 1400 °C in a typical resistive element furnace, a dark blue glass results. The reduction of Nb^{5+} ions is believed to be the source of the blue color [8], although Möncke and Ehrt [9] report that photo-induced Nb^{4+} ions (d^1 configuration) could not be identified by optical or by ESR spectroscopy, appearing to disproportionate rapidly

into Nb^{5+} (d^0) and Nb^{3+} (d^2). Rakhimov [10] analyzed ESR spectra in terms of the reaction:



And noted a shift towards the formation of Nb^{4+} pairs at lower temperatures.

In the present study, the $23\text{K}_2\text{O} \cdot 40\text{Nb}_2\text{O}_5 \cdot 37\text{P}_2\text{O}_5$ glass has been prepared and the bleaching kinetics are described using optical spectroscopy with a model based on the diffusion of oxygen and its effect on the $\text{Nb}^{4+}-\text{Nb}^{5+}$ redox reaction.

2. Experimental procedure

2.1. Glass preparation

A glass with the nominal molar composition $23\text{K}_2\text{O} \cdot 40\text{Nb}_2\text{O}_5 \cdot 37\text{P}_2\text{O}_5$, designated Nb40, was obtained by mixing the appropriate amounts of K_2CO_3 (ALDRICH, 99.5%), Nb_2O_5 (APACHE CHEMICAL INC. 99.9%), and $\text{NH}_4\text{H}_2\text{PO}_4$ (ALFA AESAR, ACS 98%) and heating the batch in an Al_2O_3 crucible at 850 °C for one hour to decompose the raw materials, and then holding the melt for one hour at 1400 °C. Nitrogen gas was bubbled into the melt through an alumina tube to improve the glass homogeneity. The melt was poured onto a steel plate, and the glass was annealed at 730 °C for 30 min before being cooled to room temperature at 2 °C/min. The resulting glass was dark blue in the center and colorless along the edges, as shown in Fig. 1. The annealing temperature was $T_g = 730$ °C, as determined by differential thermal analysis (DTA 7, Perkin Elmer) of glass powders heated at 10 °C/min in an alumina crucible under flowing N_2 . Glass compositions were

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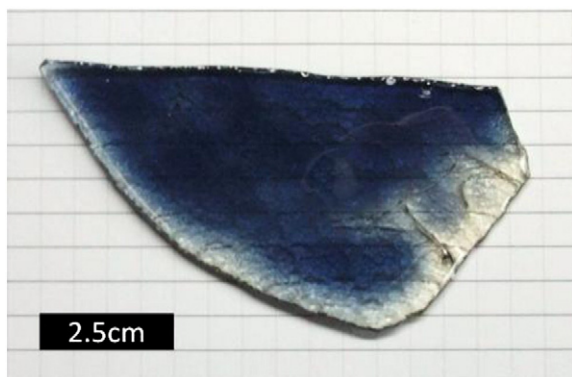


Fig. 1. Nb40 glass after annealing in air at 730 °C for 30 min and cooling to room temperature at 2 °C/min.

determined using energy dispersive spectrometry (Helios NanoLab 600 FIB/FESEM); the average of five different spots on a sample is reported.

2.2. Thermal treatments under controlled atmospheres

The annealed glass was cut into $15 \times 15 \times 1 \text{ mm}^3$ sections, polished to a mirror finish on both large faces, and then subjected to various thermal treatments. Some samples were held under flowing O_2 at 730 °C for 12 h to produce a homogeneously colorless piece of glass. These samples were then exposed to different reducing conditions, either in a silica tube furnace with either flowing forming gas (99% N_2 –1% H_2) or flowing N_2 , or in a graphite furnace with flowing Ar or flowing N_2 .

A second set of Nb40 samples were prepared to study the color bleaching kinetics. These samples were initially treated in the silica tube furnace at 730 °C for 12 h under flowing forming gas, then held in the same tube furnace under flowing O_2 , for various times and at temperatures between 650 and 730 °C. Optical spectra (THERMO SCIENTIFIC, MODEL GENESIS 10uv) were collected at room temperature from the center of each of these samples.

2.3. FTIR and ESR spectroscopies

Fourier transform infrared (FTIR) spectra were collected in transmission using a THERMO NICOLET, MODEL NEXUS 670 spectrometer, from Nb40 samples that had been exposed to both reducing and oxidizing environments. Electron spin resonance spectra were collected using a BRUKER EMX spectrometer operating in the X-band at 150 K, using liquid N_2 . Operating parameters included modulation amplitude of 3 G, microwave power of 1.968 to 1.970 mW at 20 dB attenuation, and a ~100 kHz modulation frequency. The spectra were collected with powdered samples placed in an ESR-transparent quartz glass tube.

3. Results

Table 1 shows the nominal glass composition and that measured by EDS. For these analyses, the relative cation concentrations were

Table 1
Nominal and analyzed compositions of the Nb40 glass.

Compound	Nominal (mol%)	Measured (mol%)
Al_2O_3	–	2.8 ± 0.3
K_2O	23.0	23.6 ± 0.5
P_2O_5	37.0	34.7 ± 0.3
Nb_2O_5	40.0	38.9 ± 1.0

determined and converted to the most stable respective oxides. Some Al_2O_3 was detected in the glass, transferred from the Al_2O_3 crucible and/or the Al_2O_3 bubbler tube used during melting.

Fig. 2 shows samples of Nb40 glass after 12 h at 730 °C under flowing O_2 (left) and after the same time and temperature, but under flowing forming gas (right). The dark blue color of the reduced sample was also created when initially colorless glasses were heated for 12 h at 730 °C in the graphite furnace with both nitrogen and argon atmospheres.

Fig. 3 shows the ESR spectra for these same four samples. The three blue glasses each possess ESR lineshapes which match those measured in previous studies of systems containing niobium [10]. Given such close agreement, ESR results obtained for the niobium phosphate glasses studied here presumably are due to Nb^{4+} ions occurring in binuclear pairs, as proposed by Rakhimov et al., [10]. Their proposed model is based on careful fitting of experimental ESR data and interpretation of those fits. Such fits also strongly suggest that only one type of Nb^{4+} binuclear pair occurs, due to the fact that experimental data can be predicted only for a single set of spin Hamiltonian parameters consistent with a pair of Nb^{4+} ions.

The model further predicts a shift in equilibrium from $\text{Nb}^{3+}\text{--O--Nb}^{5+}$ binuclear pairs to $\text{Nb}^{4+}\text{--O--Nb}^{4+}$ pairs as temperatures are lowered from room temperature. This shift is indicated by large increases in ESR signal intensities at temperatures below 300 K; moreover, these increases cannot be explained or predicted by simple Curie law behavior. The present work reveals similar increases, thus confirming the previous experiments and providing additional evidence for temperature-driven formation of $\text{Nb}^{4+}\text{--O--Nb}^{4+}$ binuclear pairs.

Overall, the present ESR experiments reveal substantially enhanced signal intensities in blue Nb40 glasses in comparison with colorless ones. These spectra are interpreted as evidence for the presence of ESR-sensitive Nb^{4+} ions, occurring in binuclear pairs. Generation of an increased fraction of $\text{Nb}^{4+}\text{--O--Nb}^{4+}$ pairs (versus $\text{Nb}^{3+}\text{--O--Nb}^{5+}$ pairs) at lower temperatures is confirmed here in blue Nb40 samples. The near absence of ESR signals in the colorless glass suggests that Nb^{4+} ions are almost nonexistent even at temperatures down to ~150 K.

Fig. 4 shows the optical spectra collected from Nb40 samples that were initially blue (after heat treatments in forming gas), then bleached at 730 °C in flowing O_2 for different times. With increasing time in oxygen, the broad absorption band with maximum at around 800 nm decreases in intensity.

Fig. 5 shows the FTIR spectra collected from of an Nb40 sample initially reduced in forming gas (blue in color), then heat-treated at 710 °C in flowing oxygen for different times (until colorless). The broad absorption peak centered near 2900 cm^{-1} has been assigned to hydroxyl ions in a phosphate network [11]. There is no evidence in these spectra, or in those collected from other samples, that the change in glass color is correlated with any measurable change in the residual hydroxyls in the glass.

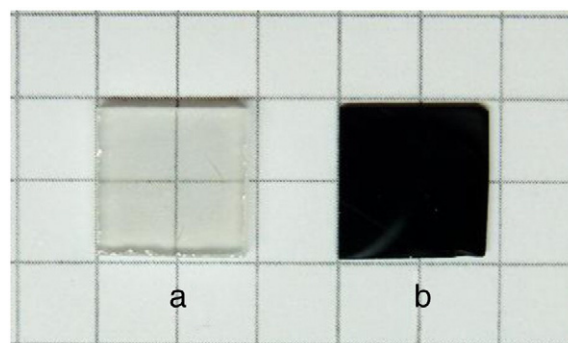


Fig. 2. Nb40 glass (a) after thermal treatment at 730 °C for 12 h under flowing O_2 , and (b) after an additional treatment at 730 °C for 12 h under forming gas.

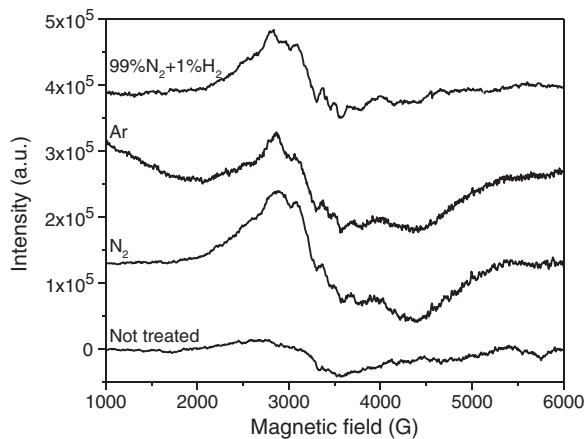


Fig. 3. ESR spectra (collected at 150 K) from samples first treated in O₂ and then held for 12 h at 730 °C in the atmosphere indicated. The sample that did not receive the second treatment was colorless and the others were dark blue.

4. Discussion

Nb40 glasses heated in oxygen near their glass transition temperature become colorless, whereas glasses heated under reducing conditions become dark blue.

ESR line shapes and magnetic fields obtained for the blue-colored niobium phosphate glasses studied here match those reported in previous studies of similar systems [10], and so the present results are interpreted as being due to Nb⁴⁺ ions occurring in binuclear pairs (Nb⁴⁺–O–Nb⁴⁺), as proposed by Rakhimov et al. In agreement with the Rakhimov model, there is an increase in the intensity of the ESR signal with a decrease in temperature, from room temperature to 150 K. The near absence of ESR signals in clear niobium phosphate glass systems suggests that Nb⁴⁺ ions are almost nonexistent even at temperatures down to ~150 K.

The presence of Nb⁴⁺ ions and their effect on the color changes may be understood by one of several possible redox reactions. For example:



Here, it is assumed that oxygen diffuses into or out of the Nb40 glass, depending on the O₂ partial pressure of the environment during the heat treatment. Bleaching of the initially reduced blue glasses can then be explained by the diffusion of oxygen into the glass to oxidize the Nb⁴⁺ ions to Nb⁵⁺ ions, as represented by Eq. (2).

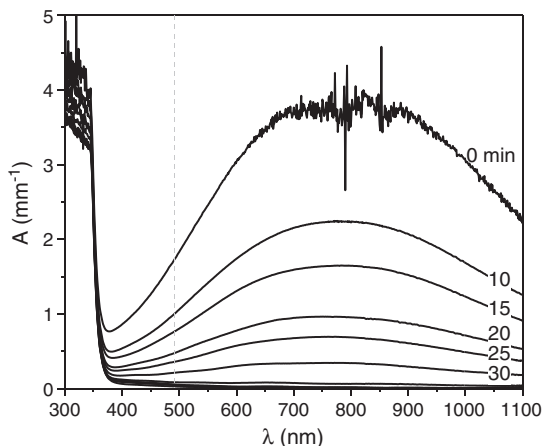


Fig. 4. UV–vis spectra for a reduced NB40 glass sample treated at 730 °C in flowing O₂ for incremental times.

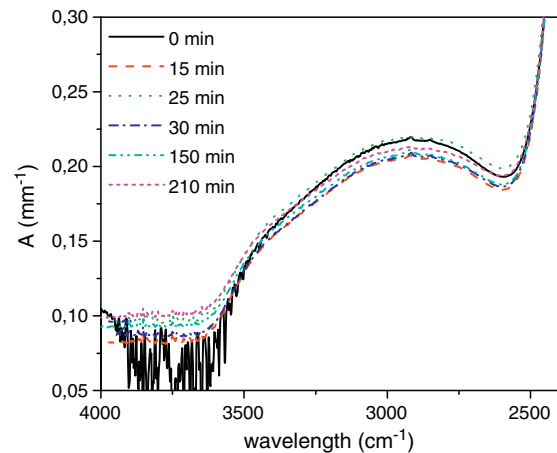
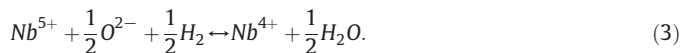
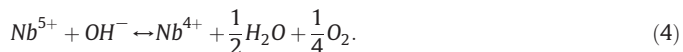


Fig. 5. FTIR spectra for a reduced Nb40 glass sample treated at 710 °C in flowing O₂ for incremental times.

A second possibility involves the reduction of Nb⁵⁺ by H₂:



Here, color is controlled by the diffusion of H₂ into the Nb40 glass under reducing conditions, or by the diffusion of water out of the glass under oxidizing conditions. A related reaction involves interactions between Nb⁵⁺ ions and residual hydroxyls in the glass:



Under low pO₂ conditions, this reaction would be driven to the right and would be dependent on the diffusion of OH⁻ ions (dehydroxylation) out of the glass.

To first test the hypothesis that the color change is controlled by a diffusion process, the optical spectra, like those in Fig. 4, were analyzed using a diffusion model. Samples can be considered infinite sheets if the amount of substance that diffuses through the edges is negligible in comparison to the amount that diffuses through the planar surfaces. This condition is satisfied when [12]:

$$\frac{a}{\sqrt{Dt}} > 10 \quad (5)$$

where a is half of the sample width (7.5 mm), D is the diffusion coefficient, and t is the time.

If the concentration of the diffusing species is initially uniform throughout the sample and equal to C_0 at time $t = 0$, and equal to C_s at the surface of the sample, then the total amount of substance (e.g., oxygen) to enter or leave a sample with thickness $2l$ after time t is given by [12]:

$$\frac{M_t}{M_\infty} = 1 - \sum_{n=0}^{\infty} \frac{8}{(2n+1)^2\pi^2} e^{-\frac{D(2n+1)^2\pi^2 t}{4l^2}}, \quad (6)$$

where M_t is the amount of the diffusing substance that entered or left the sample after time t and M_∞ is this amount after infinite time. M_t can be expressed in terms of the concentration gradient established in the sample after time t as:

$$M_t = V_{\text{sample}} \cdot \frac{1}{l} \int_0^l C(x) - C_0 dx \quad (7)$$

where $C(x)$ is the concentration gradient of the species after a given time and V_{sample} is the sample volume.

For a medium with a continuous concentration profile of an absorbing species, $C_{as}(x)$, the Beer–Lambert Law can be written as:

$$A = -\log\left(\frac{I}{I_0}\right) = \varepsilon \int_0^l C_{as}(x) dx \quad (8a)$$

and

$$(A - A_0) = \varepsilon \int_0^l C_{as}(x) - C_{as,0} dx, \quad (8b)$$

where A is the absorbance of the sample and ε is the extinction coefficient. The variations in the concentrations of the diffusing and the absorbing species C and C_{as} (the terms inside the integrals of Eqs. (7) and (8b)) are assumed to be proportional to each other; i.e. the $\text{Nb}^{4+}/\text{Nb}^{5+}$ redox is determined by the diffusion of species like those described in Eqs. 2–4. The relationship between M_t and A then becomes:

$$\frac{M_t}{M_\infty} = \frac{A(t) - A_0}{A_\infty - A_0}. \quad (9)$$

Combining Eqs. (6) and (9) leads to the following equation which describes the time dependence of the absorption data in terms of a diffusion coefficient:

$$\frac{A(t) - A_\infty}{A_0 - A_\infty} = \sum_{n=0}^{\infty} \frac{8}{(2n+1)^2 \pi^2} e^{-\frac{D(2n+1)^2 \pi^2 t}{4l^2}}, \quad (10)$$

where $A(t)$ is the absorbance after a thermal treatment for time t , A_0 is taken as the absorbance of each sample after the initial heat treatment under the reducing conditions, and A_∞ as the absorbance of the colorless sample heat treated under oxidizing conditions for a sufficiently long time.

Absorbance at 492 nm (Fig. 4) was used to calculate the “relative absorbance”, $\frac{A(t) - A_\infty}{A_0 - A_\infty}$, of samples after different times and temperatures under oxidizing conditions. This wavelength was selected because a measurable absorbance could be obtained on all samples used in this study, within the limits of detection of the spectrometer used. The “relative absorbance”, $\frac{A(t) - A_\infty}{A_0 - A_\infty}$, is plotted in Fig. 6 for samples treated in oxygen for different temperatures and times. The lines shown are the result of fitting Eq. 10, using only the diffusion coefficient, D , as a fitting parameter.

The diffusion coefficients at each temperature, determined from Fig. 6, are shown in Fig. 7. From the slope of this plot, the activation energy for diffusion was determined to be 150 ± 31 kJ/mol.

Bleaching of color from the Nb40 glass under oxidizing conditions is assumed to be due to the diffusion of oxygen into the glass where it oxidizes Nb^{4+} ions to form Nb^{5+} ions (Eq. 2). Similar color bleaching

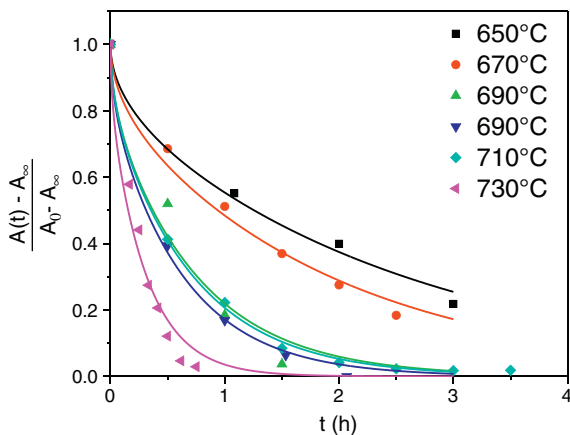


Fig. 6. Relative absorbance at 492 nm for initially blue Nb40 glasses heated in flowing oxygen for different times and temperatures.

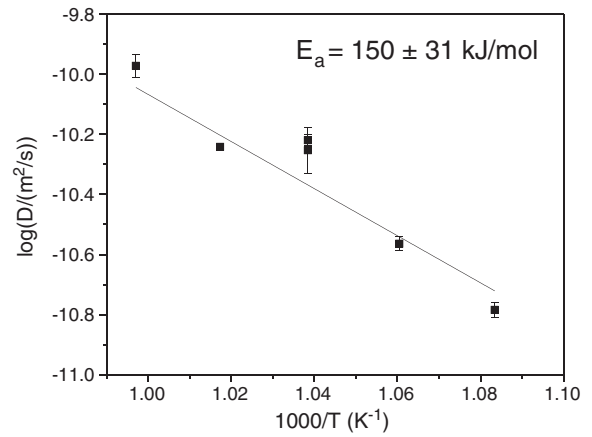


Fig. 7. Diffusion coefficients obtained from bleaching experiments (Fig. 6) for initially blue Nb40 glasses heated in flowing oxygen for different times and temperatures.

of a dark blue alkali tungsten niobium phosphate glass was noted by Tawarayama et al. [6]. However, they ascribed the color changes in terms of interactions between tungsten ions and protons from hydroxyl groups in the glass, a mechanism similar to that shown in Eq. 4. They proposed that hydrogen diffuses in and out of their glasses, controlling the concentrations of (W^{5+}/H^+) pairs that then determine the glass color. They report activation energies for color bleaching of 66 kJ/mol and 95 kJ/mol, depending on the residual $[\text{OH}]$ content of the glass.

In contrast to the findings of Tawarayama et al., the diffusion of oxygen, not hydrogen, is assumed to be the controlling step in the bleaching of the Nb40 glasses. As shown in Fig. 5, there are no changes in the infrared spectra of these glasses as they are bleached from blue to colorless, and thus there are no measurable changes in the relatively hydroxyl concentrations of the glasses. That said, the intensity of the ESR signal from the blue glasses indicates that small concentrations (ppm range) of Nb^{4+} ions are present in these reduced glasses and so it is possible that the expected small changes in hydroxyl content could not be resolved in the IR data. It is worth noting that Nb^{4+} ions are formed in glasses heated in sufficiently reducing conditions (graphite furnace under argon or nitrogen) as effectively as those under flowing forming gas. This makes it unlikely that hydrogen is needed to reduce the Nb^{5+} ions (Eq. 3). The greater activation energy for color bleaching determined in this study, compared to that reported by Tawarayama et al., is consistent with a different mechanism for the two systems: color controlled by oxygen diffusion in the present glasses and hydrogen/hydroxyl diffusion for the Tawarayama glasses. The presence of tungsten in the glasses studied by Tawarayama et al. may make them more susceptible to interactions with hydrogen than the glasses studied here, although the diffusion coefficients (and activation energies) may also be affected by the compositional differences.

Several experiments could be done in the future to help refine the mechanism for coloration. Coloration (colorless to blue) kinetics could be analyzed and compared with the decoloration kinetics reported here. If there are similar diffusion coefficients and activation energies associated with the forward and back reactions, particularly if the reduction reactions were done in H-free conditions, then this might support the oxygen-diffusion mechanism (Eq. 2). Experiments could also be done on samples with different residual hydroxyl concentrations to better understand the role of these species in the bleaching process.

5. Conclusions

The reversible color changes observed in $23\text{K}_2\text{O} \cdot 40\text{Nb}_2\text{O}_5 \cdot 37\text{P}_2\text{O}_5$ glass are attributed to changes in the oxidation state of Nb ions and are controlled by a diffusing species at temperatures near T_g . Electron spin resonance confirms that Nb^{4+} ions are responsible for the blue

color. The diffusion coefficient was calculated from systematic changes in the optical spectra of the glasses after heat treatments at different temperatures and an activation energy of 150 ± 31 kJ/mol was determined. Although oxygen diffusion readily explains these color changes, dehydroxylation reactions may also play a significant role.

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